CardioNova Health

Smarter Insights. Healthier Hearts.

AI-assisted cardiovascular risk screening | Educational demonstration project

REPORT NUMBER
CN-261105-869FBB57

CARDIOVASCULAR RISK ASSESSMENT REPORT

PATIENT NAME	ASSESSMENT DATE
Sample patient (invented)	05 Nov 2026, 10:30
AGE / SEX	**REPORT STATUS**
58 years / Male	Generated automatically; not reviewed by a clinician
ASSESSMENT TYPE	**MODEL**
Machine-learning risk screening from 13 clinical inputs	Gradient Boosting (calibrated)

ESTIMATED RISK OF HEART DISEASE

HIGH

94.2%

Risk meter (project-defined bands: below 30% low, 30% to below 60% moderate, 60% or more high)

Result reliability: **GOOD**. Estimates from the model's internal versions range from 89.9% to 97.3%.

Summary of findings

The model estimates a high probability of heart disease (94.2%) for the values entered.

- Chest pain type (Asymptomatic) raised the estimate by about 16.6 percentage points compared with a typical patient in the training data.
- Major vessels coloured (1) raised the estimate by about 12.5 percentage points compared with a typical patient in the training data.
- Thal test (Reversible defect) raised the estimate by about 9.9 percentage points compared with a typical patient in the training data.

Guidance: This sample was built from invented values. Discuss real results with a qualified clinician.

This report presents the output of a statistical model. It does not confirm or exclude heart disease and must not be used to make treatment decisions.

Factors influencing the estimate

Each bar shows how many percentage points the estimate changes when that input is replaced by the typical value in the training data. These are model associations, not medical causes, and need not add up to the total.

Clinical parameters entered

PARAMETER	RESULT	UNIT	GENERAL GUIDE (adults)	STATUS
PATIENT				
Age	**58**	years	Risk generally rises with age	-
Sex	**Male**		Reported as entered	-
SYMPTOMS				
Chest pain type	**Asymptomatic**		Interpreted by a clinician	-
Exercise-induced angina	**Yes**		No chest pain on exercise expected	**Present**
RESTING MEASUREMENTS				
Resting blood pressure (systolic)	**140**	mm Hg	Under 120; 120-129 borderline; 130+ high	**Above guide**
Serum cholesterol	**260**	mg/dl	Under 200; 200-239 borderline; 240+ high	**Above guide**
Fasting blood sugar > 120 mg/dl	**No**		Fasting glucose under 100 mg/dl is usual	**Within guide**
Resting ECG	**ST-T wave abnormality**		Normal expected	**Review by clinician**
EXERCISE STRESS TEST				
Maximum heart rate achieved	**130**	bpm	Age-predicted maximum about 162 bpm	**Below age guide**
ST depression (exercise)	**2**	mm	Under 1.0 mm is usual	**Above guide**
Slope of peak exercise ST segment	**Flat**		Upsloping is usual	**Flat**
IMAGING / LABORATORY				
Major vessels (fluoroscopy)	**1**		None (0) is usual	**Present**
Thalassemia (thal) test	**Reversible defect**		Normal is usual	**Defect reported**

Guide values are general adult reference points shown for context. They are not personal targets and not a diagnosis; categorical inputs have no numeric range.

Suggested next steps

- Share this report with a qualified healthcare professional and discuss it together with your symptoms, examination and tests.
- **If you have chest pain, severe breathlessness, fainting or any emergency symptom, seek emergency care immediately.** Do not wait for a report.
- Ask: What does an estimate like this mean for someone with my history and symptoms?
- Ask: Which of my test results (ECG, stress test, imaging) should be repeated or explained in more detail?
- Ask: Do I need further tests or a follow-up, and how soon?

CLINICIAN REVIEW (OPTIONAL)			
Reviewed by (name)		**Registration no.**	
Signature		**Date**	
Comments			

LIMITATIONS AND MEDICAL DISCLAIMER

About the method. Model: Gradient Boosting, trained on 297 patient records from the UCI Cleveland dataset. Held-out test ROC-AUC 0.925 (95% interval 0.851 to 0.982). The small, single-site, mostly older training population limits how well the model fits other people.

CardioNova Health is an academic machine-learning project. This report is for educational and demonstration purposes only; it is not a hospital-issued diagnostic report, does not confirm or exclude heart disease, and must not be used to make treatment decisions. The model and its probabilities have not been clinically validated for individual patient care. Consult a qualified healthcare professional for medical assessment and interpretation of clinical tests.

Project-defined risk bands: below 30% = Low; 30% to below 60% = Moderate; 60% or above = High. These thresholds are not official clinical risk classifications.